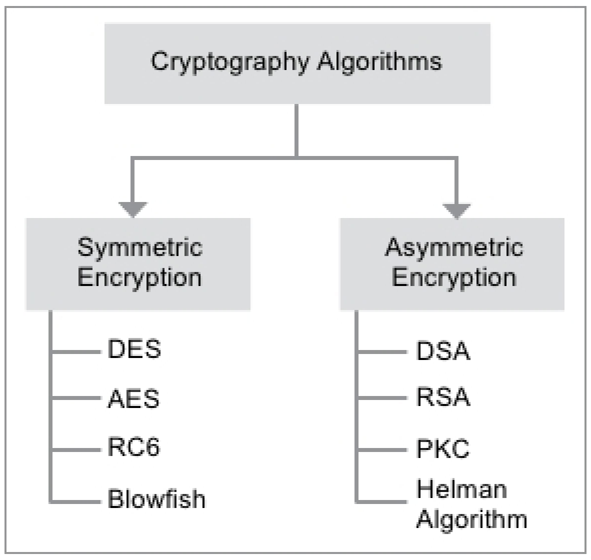

Cryptographic algorithms (Credit: https://link.springer.com)

Testing before manufacturing should still be preferred as micro-codes can decrease processor efficiency.

Wearable and IoT devices

As devices continue to become smart, kits and development boards for the Internet of Things (IoT) applications are getting popular. There are numerous hardware development platforms with features required for IoT devices.

Different IoT boards are available in the market for different applications such as smart locks and thermostats. Popular ones include Espressif ESP8266 boards, Particle.io, as well as others from Intel and Adafruit.

If the device is a wearable, it should be made comfortable and have no harmful side-effects. Along with the MCU, which is central to the IoT, boards come with inbuilt sensors and other components for physical computing. Sensors have to be highly accurate, especially when it comes to industrial IoT (IIoT) systems.

PCB designing is foremost when it comes to IoT products. Conventional PCBs cannot accommodate the ever-increasing demands in these devices. Flex and high-density interconnect (HDI) PCBs are ideal to build devices smaller in size with maximum utilisation of board space. These also provide low power consumption throughout the lifetime of products, which is important since IoT devices have to continuously communicate with the network.

These approaches ensure high performance even in harsh environments. Proper grounding and stack-up reduce issues related to thermal and power dissipation. Frequent testing decreases chances of error later on. Multi-chip modules (MCM) are used to connect multiple 3D ICs on a single die.

Since IoT devices remotely control connected devices, it is important to choose the right connectivity standard in the initial stage itself to meet the range requirements for a particular application. It is better to reuse tools that have already been tested for other similar applications rather than building the product from scratch.

Using fewer but effective components reduces manufacturing cost. Simulation software like PSpice are useful for testing the designs before building the actual hardware. With a large number of connected devices in the IoT environment, it is essential to include hardware-based protection besides software to protect against IP theft, differential power analysis (DPA) attacks, reverse engineering attacks and the like.

Network security is important as data communication with the network, which is usually on cloud, takes place over Wi-Fi. Steve Quane, executive vice president - network defense and hybrid cloud security, Trend Micro, says, "The IoT market will continue to grow, especially with landscape changes like 5G. Enterprises must be ready to protect their Industry 4.0 environments."

Each component selected for the board should meet necessary compliances and certifications like Restriction of Hazardous Substances (RoHS) Directive and EMC/EMI compliance.

Indigenisation

Developing microprocessors and MCUs indigenously is essential to reduce the dependence on imports and make India an electronics design and manufacturing hub. This year, students from IIT-Bombay developed a medium-sized microprocessor, called AJIT, that can perform basic operations. Vendors can buy a licence to get the processor design, while software tools can be accessed freely by everyone.

In 2018, RISECREEK chips were developed by IIT-Madras for use in computing devices and embedded systems. As technology continues to evolve, designers need to stay updated to reduce cost, board space and design time for their projects. EFY